An important advantage of island networks, in addition to their lower-dimensional representation, is that, combined with this ranking assumption, they enable us to model the degree of *homophily*—the extent to which an individual interacts with others that have common characteristics as herself. Common characteristics for us are those that are relevant for prior beliefs, and therefore, by construction, individuals have more in common with those on the same island as themselves. As a result, homophily will be higher when most links are within islands and links between islands are sparse (high p_s and low p_d).[14]

Second, we shift our focus from *uniformly more sharing* to the weaker notion of *content virality* (which we define as more sharing, but not necessarily uniformly so). The reason for this is that, while our game continues to exhibit strategic complementarities, changes in the sharing network do not induce monotone shifts, and thus the most and least sharing equilibria do not necessarily shift in the same direction. We will see, nevertheless, that content virality will change in well behaved ways.

Formally, we define content virality as follows. We suppose that an article is seeded at some agent i^* at $t = 1$. We then define $\mathbf{S}_{i^*}$ as the (random) proportion of the population that shares when agent i^* is the seed agent in the most-sharing equilibrium σ. We say σ_1 has more *content virality* than σ_2 if $\max_{i^*} \mathbb{E}_{\sigma_1}[\mathbf{S}_{i^*}] \geq \max_{i^*} \mathbb{E}_{\sigma_2}[\mathbf{S}_{i^*}]$. In words, our notion of content virality compares the spread of an article provided that it starts from the seed that is most favorable to its ultimate circulation. The reason we start from the most favorable seed is that, as we will see in the next section, social media platforms have an incentive to implement sharing algorithms that place articles in such favorable seeds. For future reference, we also note that content virality is also the same as expected overall engagement with an article, conditional on favorable seeding.

Finally, we define homophily more formally. We say that an island network with (p_s, p_d) has *more homophily* than an island network with (p'_s, p'_d) and the same expected degree of each agent, but where $p_s > p'_s$ and $p_d < p'_d$.[15] From Theorem 1, We know that the equilibrium is in cutoff strategies of the form $(\mathbf{b}^*, \mathbf{b}^{**}) \equiv (b^*_1, b^{**}_1, \ldots, b^*_N, b^{**}_N)$. However, because there is symmetry within islands, equilibria now take a simpler, "semi-symmetric" form as shown in the next lemma.

Lemma 1. *All equilibria are semi-symmetric: for every equilibrium, there exist $\{(b^*_\ell, b^{**}_\ell)\}_{\ell=1}^k$ such that $b^*_i = b^*_{\ell_i}$ and $b^{**}_i = b^{**}_{\ell_i}$ for all agents i in island ℓ.*

The simplification established in Lemma 1 will allow us to work with a lower dimensional cutoff vector (just two cutoffs for each island).

4.1 Comparative Statics: Homophily

The next theorem, characterizing the effects of homophily on the spread of misinformation, is our first main result:

Theorem 2. *There exist $0 < \underline{r} < \bar{r} < 1$ such that:*

[14] The homophilic structure and greater congruence of beliefs within islands are consistent with the evidence presented in Bakshy et al. (2015): "friend networks" on Facebook are ideologically segregated, with the median share of friends from the opposing ideology around only 20%. Mosleh et al. (2021b) provides evidence of similar homophily on Twitter.

[15] Because network density raises connectivity and can directly increase virality, we hold network density fixed in order to isolate the effects from homophily.